

Generation Interconnection Request System Impact Study Report

For Queue Project Q24-2392

Luka Tap 138 kV
September 19, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2392. The Interconnection Customer proposes a 100.0 MW solar photovoltaic facility interconnecting at the Luka Tap 138 kV substation in the Tennessee area. The request was received on March 19, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 394 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

This project does not cause any thermal violations under system intact or single contingency conditions. Estimated network upgrade cost: \$0

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

No overloads identified.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

None. Total network upgrade cost allocated to this project: \$0.

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.